

Stopping Target Geometry Optimization for Run 1A CE

Autoresearch loop: LLM-greedy $\rightarrow$ Bayesian Optimization

O. Oksuzian

Mu2e

May 4, 2026

Goal and metric

Maximize **Run 1A Coherent Electron** $S/\sqrt{B}$ by varying the muon stopping-target geometry.

- ▶ Single scalar metric: $S/\sqrt{B}$ from `rough_run1a_sensitivity.C`
- ▶ Per-config window scan picks optimal $[x_1, x_2]$ in p -space
- ▶ Cosmic background is *hard-coded* in the analysis macro ($\sim 1.8 \times 10^{-3}$ ev/s/MeV/c)
- ▶ DIO $\ll$ cosmic above 100 MeV/c $\Rightarrow$ optimum is essentially *signal acceptance / window-width tradeoff*

Running-period assumption (in `rough_run1a_sensitivity.C`)

All $S/\sqrt{B}$ values are computed at a *fixed* hypothetical exposure:

Quantity	Value	Note
$N(\text{POT})$	10^{18}	assumed exposure
Beam mode	1BB	1.6×10^7 POT/event
Simulated events	6.25×10^{10}	derived
On-spill seconds	$\sim 1.06 \times 10^5$ s	~ 29.4 h beam-on
$R_{\mu e}$ assumption	10^{-9}	signal hypothesis
$\text{BR}(\mu\text{-capture, Al})$	0.609	for signal branching
Cosmic rate (hard-coded)	$\sim 1.8 \times 10^{-3}$ /s/MeV/c	not from sim

Context: 10^{18} POT is $\sim 10\times$ actual Run 1A exposure. The number is a normalization that scales signal/DIO/cosmic the same way; only ratios across configs matter for the optimization.

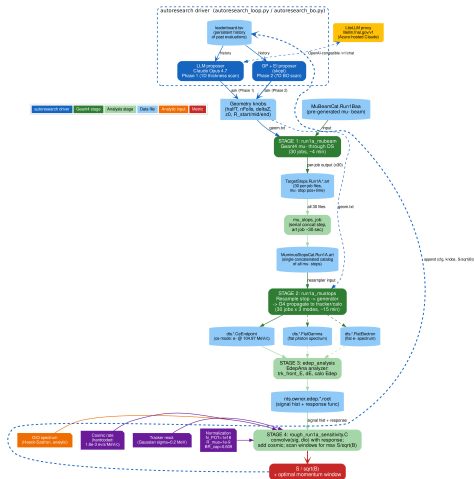
To project to real Run 1A: multiply $S/\sqrt{B}$ by $\sqrt{N_{\text{actual}}/10^{18}} \approx 0.32$. $\Rightarrow$ baseline 3.31 $\rightarrow \sim 1.0$; optimum 3.81 $\rightarrow \sim 1.2$ at real Run 1A.

Per-evaluation pipeline (~ 20 min on 30 cores)

1. Fork from `config_v00` (37-foil baseline) via `new_config.sh`
2. Override target geometry in `config_<X>/run1a_beam/geom.txt`
3. Stage `run1a_mubeam`: 30 jobs $\times$ 5 k events (G4 muon beam $\rightarrow$ TargetStops)
4. Stage `run1a_mustops`: 30 jobs $\times$ 10 k events, modes `{ce, flat_gamma, flat_electron}`
5. Read $S/\sqrt{B}$ from `rough_run1a_sensitivity.log`

Workdir setup: prepend our tree to `FHICL_FILE_PATH` and `MU2E_SEARCH_PATH`; reuse `mmackenz/run1b/` for the `czMin`-patched Offline source and pre-built `EdepAna.module.so`.

Autoresearch pipeline: outer loop + per-evaluation data flow



Top: **autoresearch driver** (dashed cluster) – leaderboard + selectable proposer (LLM via LiteLLM in Phase 1, GP+EI BO in Phase 2). Middle: Geant4 stages 1–2. Bottom: analysis stages 3–4. Loop closes via **leaderboard.tsv**.

Key physics points and what BO moves

The CE generator is “triggered.” Every μ^- stop emits an e^- at endpoint kinematics; we don't wait for real $\mu \rightarrow e$ conversion ($\text{BR} \sim 10^{-15}$). The macro scales this artificial population by $R_{\mu e} = 10^{-9}$ and $N(\text{POT}) = 10^{18}$ to get an absolute event count.

Stages 1 & 2 are decoupled via MuminusStopsCat: mubeam runs once per geometry; mustops resamples the catalog 3 times (one per generator mode).

Cosmic and DIO never touch Geant4.

- ▶ Cosmic: hardcoded flat distribution $\times$ assumed rate
- ▶ DIO: Heeck-Szafron analytic spectrum, convolved with response

$\Rightarrow$ all geometry-driven gains move only through *signal acceptance* $\div$ *window width*.

What the BO actually moves:

geometry knobs $\rightarrow$ (sig_eff, signal-spectrum shape) $\rightarrow S/\sqrt{B}$

- ▶ halfT, nFoil, deltaZ, z0 – change *how many* μ^- stop and *where*
- ▶ R_start, R_mid, R_end – change *which* μ^- stop (off-axis acceptance)
- ▶ All knobs feed Stages 1+2; Stages 3+4 are deterministic post-processing

Baselines

Three runs of the default 37-foil target ($t_{1/2} = 0.0528$ mm):

Run	Window (MeV/c)	$S/\sqrt{B}$
Mike's config_v00 (Apr 18)	[104.5, 104.5]	3.35
Our config_v24 (yesterday)	[103.9, 104.7]	3.34
Our config_v00 (today)	[103.9, 104.7]	3.31

Cross-validates the pipeline. All three within $\sim 1\%$ (MC noise).

Phase 1: 1D scan over foil half-thickness (LLM-greedy)

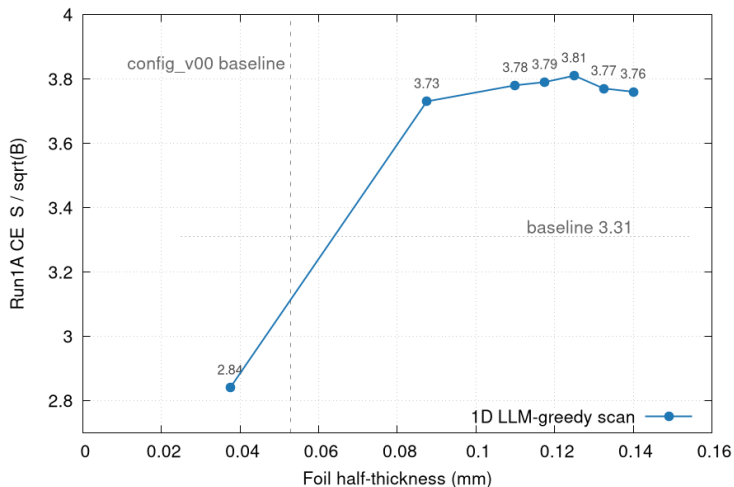
Single knob: `stoppingTarget.halfThicknesses` $\in [0.025, 0.150]$ mm.

Loop (10 iterations, ~ 3.5 h total):

1. Read leaderboard (history)
2. LLM (azure/claude-opus-4-7 via `litellm.fnal.gov/v1`) proposes next t
3. Fork config, write geom override, run pipeline (~ 20 min)
4. Parse $S/\sqrt{B}$, append to leaderboard

LLM behavior: 3-point spread to characterize, then bisect around best.

Phase 1 results: 7 successful evaluations



Best: $t_{1/2} = 0.125$ mm, $S/\sqrt{B} = 3.81$ (+15% vs. baseline 3.31)

Plateau: $S/\sqrt{B} \in [3.76, 3.81]$ for $t_{1/2} \in [0.110, 0.140]$ mm

Why thickness has an optimum

Two competing effects:

- ▶ Thicker foils $\Rightarrow$ more μ^- stops $\Rightarrow$ *linear* signal gain
- ▶ Thicker foils $\Rightarrow$ more E -loss + multiple scattering before tracker $\Rightarrow$ broader CE peak $\Rightarrow$ optimal window widens $\Rightarrow$ more cosmic bkg

$t_{1/2}$ (mm)	Window width (MeV/c)	Signal	Cosmic
0.0375	0.8	39	190
0.0875	1.2	61	270
0.1250	1.8	75	390
0.1400	2.0	77	420

0.0375 $\rightarrow$ 0.125 mm: signal $\times 1.9$, cosmic $\times 2.0$

$\Rightarrow S/\sqrt{B} \times 1.34$ (matches: 2.84 $\rightarrow$ 3.81)

Phase 2: 7D Bayesian Optimization

Knob set:

Knob	Range	Type
halfT	[0.025, 0.150] mm	continuous
nFoils	[20, 60]	integer
deltaZ	[10, 30] mm	continuous
z0	[5700, 6200] mm	continuous
R_start	[50, 100] mm	cont. (foil 0)
R_mid	[50, 100] mm	cont. (foil $N/2$)
R_end	[50, 100] mm	cont. (foil $N-1$)

Outer-radius profile: quadratic Lagrange interpolation through 3 control points $\Rightarrow$ smooth taper / bulge / inverse-taper with only 3 BO knobs.

Constraints: $z_0 + (n_{\text{foils}} - 1) \Delta z \leq 6500$ mm; `stoppingTarget.holeRadius` fixed at 21.5 mm.

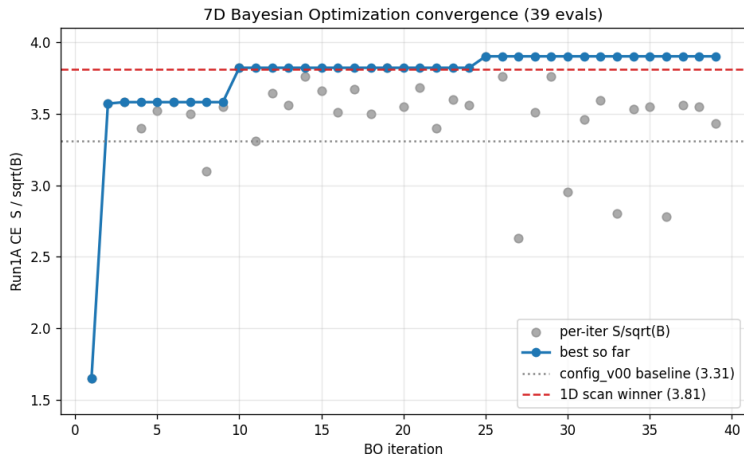
BO setup

- ▶ `scikit-optimize` GP regressor + Expected Improvement acquisition
- ▶ Seeded with the 7 thickness priors $\Rightarrow$ GP starts with data along one axis
- ▶ Budget: **40 fresh evaluations**, completed in ~ 15 hours wall-clock
- ▶ Driver: `autoresearch_bo.py`, resumable from `leaderboard_bo.tsv`
- ▶ Validation phase planned: top-3 candidates re-run at 50 *k* events/job

Why surrogate at this stage?

- ▶ 1D: surrogate overkill, LLM-greedy or sweep is fine
- ▶ 5D–7D: vanilla BO converges in ~ 30 –40 trials at this dimensionality
- ▶ Effective dim is low: physics smoothness encoded by the quadratic basis

Phase 2 results: convergence



Best: config_bo025, $S/\sqrt{B} = 3.90$ (+18% vs. baseline, +2.4% vs. 1D winner)

Iter 1: extreme corner (1.65, informative failure). Iter 10: matches 1D winner. Iter 25: finds 3.90. Iters 26–40: plateau.

Phase 2 results: best configuration

config_bo025 vs. baseline:

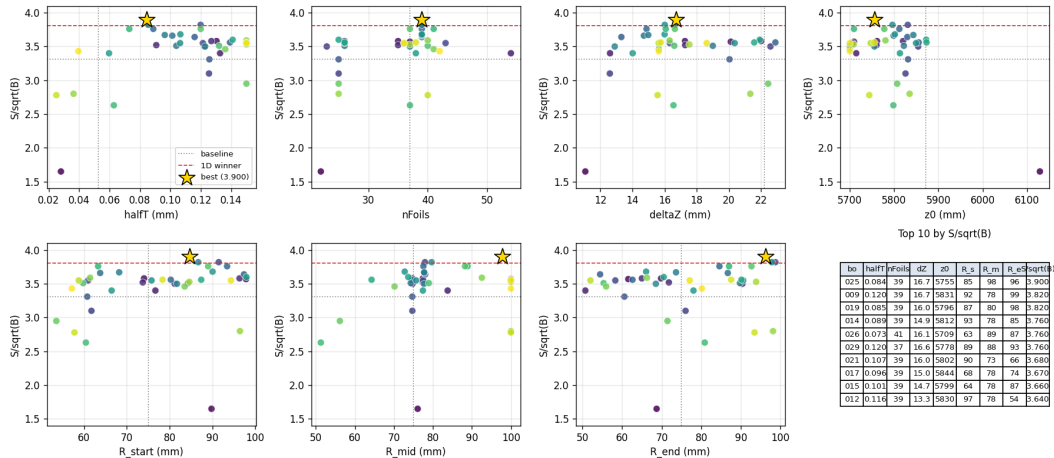
Knob	config_v00 (baseline)	config_bo025
halfT	0.0528 mm	0.0842 mm
nFoils	37	39
deltaZ	22.22 mm	16.71 mm
z0	5871 mm	5755 mm
R_start	75 mm	84.6 mm
R_mid	75 mm	97.8 mm
R_end	75 mm	96.3 mm
$S/\sqrt{B}$	3.31	3.90 (+18%)

Top 3 (within MC noise of each other):

bo025 (3.90), bo009 (3.82), bo019 (3.82) – statistically tied.

Phase 2 results: per-knob projections

7D BO: 1D projections of $S/\sqrt{B}$ per knob (39 evals, color = iteration)



Color: BO iteration. Star: best ($S/\sqrt{B} = 3.90$). Dotted vertical: baseline. Dashed horizontal: 1D-scan winner (3.81).

Which knobs actually mattered?

After-the-fact regression on 39 evaluations (excluding the 1.65 corner outlier):

Knob	R^2 (linear, alone)	top-10 range used	verdict
R_start	0.155	68% of [50,100]	weak constraint
halfT	0.115	38% of [0.025,0.150]	moderate
nFoils	0.059	10% of [20,60]	strongly converged (37–41)
R_mid	0.032	50%	weak
deltaZ	0.012	17% of [10,30]	strongly converged (13–17)
z0	0.002	27%	moderate
R_end	0.000	89%	essentially flat

Linear fit, all 7 knobs together: $R^2 = 0.42$. The other 58% is nonlinear / coupling / MC noise.

Key takeaway: 2 of 7 knobs (nFoils, deltaZ) are tightly converged; the radius profile (R_*) contributes weakly. R_end can be dropped from any future scan – it is genuinely unconstrained, not under-sampled.

Why more 7D iterations won't help

Three independent reasons combined:

1. **Physics ceiling.** $S/\sqrt{B} = \text{signal} / \sqrt{\text{cosmic}}$, with cosmic hard-coded. Geometry can only improve signal acceptance / window width; the 1D scan already approached the ceiling.
2. **Noise floor.** Per-eval noise $\sim 2\%$ on $S/\sqrt{B}$. Top-3 spread (3.90, 3.82, 3.82) is at noise. Iter 41 returning 3.95 would be statistically indistinguishable from 3.90.
3. **BO has converged.** 15 iterations after iter 25 produced no improvement. Late iterations sampled known corners (e.g. `R_mid=100`, `halfT=0.15`) returning ~ 3.55 , well below 3.90.

Slice test (configs with `nFoils` $\in [37, 41]$ and `deltaZ` $\in [14, 18]$, $N = 17$): $S/\sqrt{B}$ mean = 3.55, max = 3.90. The remaining 5 knobs vary widely within this slice but the maximum is still 3.90.

$\Rightarrow$ *This 7D box has been effectively explored.* Real gains require changing the problem – new dimensions or different analysis.

Caveats

- ▶ **Cosmic background is hard-coded** in the macro. All gains move through (signal acceptance) $\div$ (window width) – not from cosmic modeling improvements.
- ▶ **MC noise** per evaluation $\sim 1\%$ – 2% ($30 \text{ jobs} \times 10 \text{ k events}$). Top points within $\sim 1.5\%$ are statistically tied.
- ▶ **TSdA = polyethylene neutron absorber** in `config_v00`, not the Run 1B Al-endplate variant. Adding the endplate knobs would be a separate scan (+5 dims).
- ▶ **Run 1B physics not in this study** – field-off Run 1B stages take $\sim 44 \text{ min/eval}$ and would change the iteration economics.

Available knobs beyond the 7D scan

The Mu2e geometry code (`StoppingTargetMaker.cc`) supports per-foil specifications *straight from the geom.txt file* – no Offline source changes:

Parameter	Vector type	Currently in BO?
<code>radii</code>	per-foil outer R	yes (3 knobs, quadratic)
<code>halfThicknesses</code>	per-foil thickness	1 scalar; could go quadratic (3)
<code>zVars</code>	per-foil z offset	not used; non-uniform spacing (3)
<code>xCos</code>	per-foil tilt (about y)	not used (1 uniform or 3 quad.)
<code>yCos</code>	per-foil tilt (about x)	redundant (φ -symmetric)
<code>xVars, yVars</code>	per-foil x,y offset	transverse misalignment
<code>foilTarget_support*</code>	support wires	material / count near foils

Pattern: 3 control points $\rightarrow$ quadratic Lagrange $\rightarrow$ smooth per-foil profile. Same 3-knob cost as `R_start`, `R_mid`, `R_end`.

Full smooth-target parametrization (theoretical)

If we promote every per-foil quantity to a quadratic profile:

Knob group	BO knobs
Foil thickness profile (3 control points)	3
Foil outer radius profile (3 control points)	3
Foil spacing profile (3 control points)	3
Foil tilt profile (3 control points, x only)	3
nFoils	1
z0 (target center)	1
Total	14D

Why y-tilt is excluded: Mu2e is approximately φ -symmetric (beam, solenoid field, tracker, calorimeter all centered on z). x- and y-tilt produce identical $S/\sqrt{B}$ up to a 90° rotation; adding yCos as a BO knob just explores a redundant circle in (xCos, yCos) space.

14D budget: vanilla GP-El BO needs ~ 80 – 120 trials. At 20 min/eval that's ~ 30 hours. Practical: run cheap 1D probes at the current 7D winner to triage which extra knobs deserve to graduate to BO dimensions.

Cheap 1D probes (recommended path)

After the 7D scan converges, run 1D scans at the current best to decide which knobs are worth promoting:

Probe	Cost	Decision rule
Per-foil thickness (uniform $\rightarrow$ tapered)	5 evals, 1.5 h	promote if $>2\%$ on $S/\sqrt{B}$
Per-foil spacing (uniform $\rightarrow$ tapered)	5 evals, 1.5 h	promote if $>2\%$
Foil tilt (uniform, $\theta \in [0, 30^\circ]$)	5 evals, 1.5 h	promote if $>2\%$
Support-wire radius	5 evals, 1.5 h	rarely matters; quick sanity

≈ 30 evals $\times$ 20 min = 10 hours total to triage all four. Each “promoted” knob group then becomes 3 BO dimensions in a follow-up scan.

Rationale: avoid expanding to 11D or 14D BO without evidence that the extra dims actually move the metric. Cheap to verify, expensive to ignore.

Beyond the stopping target

Knobs we deliberately did *not* include in the autoresearch scan, but worth noting:

- ▶ **TSdA Run 1B endplate** – 5 knobs (material, r4, rin, halfLength, z0). Currently polyethylene neutron absorber; switching to Al-as-stopping-target is a Run 1B physics concept.
- ▶ **Proton Absorber** (PA / IPA) – z, inner radius, thickness. Sits between target and tracker; affects CE energy loss and resolution.
- ▶ **Tracker placement** – ds2.halfLength, tracker.inDS2Vacuum.
- ▶ **DS field scale factor** – changes momentum reconstruction.
- ▶ **Stopping-target material** – non-Al would change capture branching; breaks the analysis assumptions but worth investigating for material studies.

And on the analysis side:

- ▶ **Cosmic background model** (currently hard-coded $\sim 1.8 \times 10^{-3}/\text{s/MeV/c}$). Replacing with sim-derived spectrum is the single biggest unknown for absolute $S/\sqrt{B}$.
- ▶ **Tracker resolution** (currently Gaussian $\sigma=0.2$ MeV).

Next steps

Done:

- ▶ 1D LLM-greedy thickness scan (10 evals, +15% over baseline)
- ▶ 7D BO over thickness + foil-count + spacing + z + radius profile (40 evals, +18%)

Recommended immediate:

1. **Validate top 3** (bo025, bo009, bo019) at $5\times$ stats (50 k events/job). ~ 3.75 h. Tells us whether 3.90 is real or a 2% upfluctuation.
2. **Promote winner** into a hand-vetted config_v40 once validated.

Optional follow-ups (to extract gains beyond the cosmic ceiling):

- ▶ **Drop** R_end, add per-foil *thickness* profile (3 knobs) and per-foil *spacing* profile (3 knobs) $\rightarrow$ 11D BO with new physics axes
- ▶ **Add TSdA Al-endplate knobs** $\rightarrow$ Run 1B endplate variant
- ▶ **Replace hard-coded cosmic** with sim-derived spectrum (analysis-side change)
- ▶ **Extend to Run 1B final-stage CE sensitivity** (full pipeline, ~ 44 min/eval, field-off regime)